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SEAT No. :

PE-231

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[6580]-594

B.E. (E & TC) (Insem.)

VLSI DESIGN AND TECHNOLOGY
(2019 Pattern) (Semester - VII) (404182)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagrams must be drawn whenever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume any missing data if necessary.

Q1) a) Write VHDL Code and its test bench of full adder. [8]

b) What is meant by concurrent and sequential statements in VHDL?
Describe with examples. [7]

OR

Q2) a) Explain in detail digital design flow. [8]

b) Write VHDL code for 4:1 Mux using behavioral modeling style. Also
write its test bench code. [7]

Q3) a) Differentiate following: [8]

- i) Mealy and Moore Machine
- ii) Fan In and Fan out

b) What is Clock Skew? What are techniques to minimize it? [7]

OR

Q4) a) Write short note on: [8]

- i) Metastability
- ii) Noise Margin

b) What are different signal integrity issues? How to minimize it. [7]

